ECO-4001B LET Review

LLE – Mathematics and Statistics Skills

Underground Airways is conducting a review of their customers and their own operations.

1. Data are collected to find annual income of passengers booking a business class ticket. The annual incomes of customers are known to follow a normal distribution. A sample of 16 customers were asked for their annual income:

   Annual income (thousands £):

   60.3, 46.3, 45.7, 45.9, 45.6, 38.6, 63.8, 46.5, 53.6, 37.5, 54.1, 63.9, 49.9, 58.2, 42.9, 63.2

   These data can be summarised:

Statistic	
Observations	16
Mean	
Median	48.2
Standard Deviation	8.8
Standard Error of the Mean	

   (a) Calculate the two missing values in the table.

   (b) Historically the mean annual income of business class ticket customers was £47000. Underground Airways want to test, with this sample, whether this has changed.

      i. What would the null hypothesis of this test be?

      ii. What would the alternative hypothesis of this test be?

      iii. How many degrees of freedom would you have in carrying out a t-test?

iv. Calculate the t statistic for this test.
v. From the t distribution table, what would the critical t value be at the 5% level?
vi. Would you reject the null hypothesis in this test?

(c) Construct a 95% confidence interval for the population mean of business class customers income, from this sample.

(d) Construct a 99% confidence interval for the population mean of business class customers income, from this sample

2. Underground Airways also offer a 'business plus' ticket. From a sample of customers' incomes for this ticket:

Statistic	
Observations	20
Mean	57
Median	56.5
Standard Deviation	8.5

(a) Find the 90% confidence interval for the mean

(b) A t-test is carried out to determine if the mean is significantly different from £52000. The results give:

$t(19) = 2.6307, p = .0165$

Interpret this result.

3. A sample of the Underground Airways customers were asked to rate their experience of the company through a series of questions. From these questions a satisfaction score can be calculated. A summary of the satisfaction scores by business and business plus passengers:

Statistic	Business	Business Plus
Observations	120	130
Mean	18	20
Standard Deviation	6	6.8

A t-test will be be performed to test for a significant difference between satisfaction scores for business and business plus customers.

(a) What are the null and alternative hypotheses of the two sample test?

(b) Calculate the pooled variance and pooled standard deviation statistics.

(c) Calculate the t-statistic and state the degrees of freedom.

(d) At the 5% significance level what conclusion would you draw about the difference in satisfaction rating between the two passenger groups?

4. Underground Airways record data, over a particular week, for how the tickets are booked. They offer Economy, Business, Business Plus and First Class tickets that can all be booked by visiting a travel agent, over the phone or on the internet. The number of customers for these factors can be summarised in the table:

	Economy	Business	Business Plus	First Class	TOTAL
Travel Agent	25	18	4	23	70
Phone	120	80	45	45	290
Internet	140	90	65	25	320
TOTAL	285	188	114	93	680

A customer is selected at random, in a prize draw.

(a) What is the probability that the customer will be from economy?

(b) What is the probability that the customer will have booked via the internet?

(c) What is the probability that the customer was from economy and booked via the internet?

(d) Given that the customer booked economy, what is the probability that they booked via the internet?

(e) Are the events booking economy and booking via the internet independent? Give an explanation for your answer.

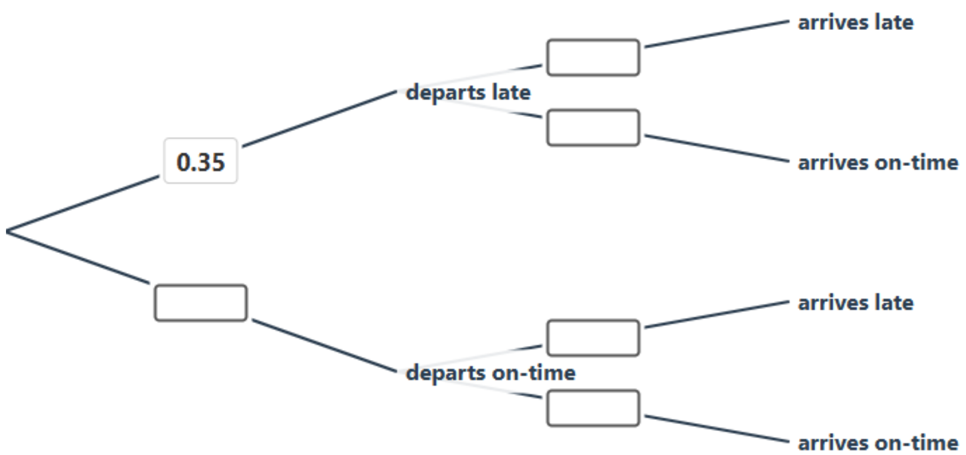
(f) What is the probability that the customer booked either business or business plus?

(g) Explain why the events booking business and booking business plus are mutually exclusive.

(h) What is the probability that the customer did not book via a travel agent given that they were booked first class?

5. Underground Airways services depart late 35% of the time. If the service departs late then they arrive late at the destination 75% of the time. Services that depart on time will arrive late at the destination 20% of the time.

(a) Complete the tree diagram



For a randomly selected flight:

(b) What is the probability that the service departs and arrives on time?

(c) What is the probability that the service arrives on time?

(d) Given that the service arrived on time, what is the probability that it departed on time?

6. Customer complaints occur during 1 in 10 services operated by Underground Airways. During a randomly chosen day 12 services are operated by Underground Airways.

   (a) What is the probability that none of the 12 services receive a complaint?
   (b) What is the probability that will be no more than 1 service receiving a complaint?
   (c) What is the probability that at least 2 services will receive a complaint?